

Multiple Choice

1. **Answer:** A

Options: A) MgO magnesium oxide; B) AgNO₃ silver nitrate; C) CuCl₂ copper(II) chloride; D) H₂SO₄ sulfuric acid; E) K₂Cr₂O₇ potassium dichromate

Note: MgO is incorrectly named magnesium oxide because it should be referred to as magnesium(II) oxide to indicate the oxidation state of magnesium in the compound.

2. **Answer:** A

Options: A) Bent; B) Trigonal pyramid; C) T-shaped; D) Trigonal planar; E) Trigonal bipyramidal

Note: The correct answer is "Bent" because a molecular geometry with five sigma bonds and one nonbonding pair of electrons results in a distorted tetrahedral arrangement, leading to a bent shape due to the repulsion between the lone pair and bonding pairs.

3. **Answer:** D

Options: A) $3 \times 10^{-23}g$; B) $7.15 \times 10^{-22}g$; C) $1.38 \times 10^{-21}g$; D) $6.41 \times 10^{-22}g$; E) $8.41 \times 10^{-22}g$

Note: The mass of one molecule of cholesterol is calculated by dividing its molecular mass (386 g/mol) by Avogadro's number (6.022 x 10²³ molecules/mol), resulting in a mass of approximately 6.41 x 10⁻²² g per molecule.

3. **Answer:** A

Options: A) 5/2, 4/3; B) -5/3, 4/2; C) 5/3, -4/2; D) 2, -10; E) 4, -5

Note: The correct answer, 5/2 and 4/3, results from applying the quadratic formula to the equation $6x^2 - 7x - 20 = 0$, which reveals the solutions to the values of x where the quadratic expression equals zero.

4. **Answer:** A

Options: A) 21 atm; B) 28 atm; C) 30 atm; D) 12 atm; E) 18 atm

Note: The correct answer of 21 atm is derived from the ideal gas law, $PV = nRT$, where the number of moles (1.0 mol), the universal gas constant (0.0821 L · atm/(K · mol)), and the temperature (298 K) yield a pressure of approximately 21 atm when the volume is 1.0 dm³.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The resulting molarity of the sucrose solution after mixing is 0.33 M because the total number of moles of sucrose divided by the total volume of the solution in liters equals 0.33 M.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the nuclide mass of N^{13} is accurately measured at 13.00335 amu, which accounts for the mass defect resulting from the binding energy in the nucleus after the beta-plus decay process, where a proton is converted into a neutron, emitting a positron and a neutrino.

8. **Answer:** True

Options: A) True; B) False

Note: The pH of a 0.001 M aqueous solution of KOH is 13 because KOH fully dissociates in water, resulting in a hydroxide ion concentration of 0.001 M, which corresponds to a pH calculated as $14 - (-\log[0.001]) = 13$, confirming it as a strong base.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the work function of potassium is a measure of the minimum energy required to remove an electron from the metal's surface, and it cannot be lower than the threshold energy, which is defined as the energy required to initiate the photoelectric effect.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the balanced chemical reaction for the combustion of carbon monoxide (CO) shows that one mole of CO reacts with half a mole of O_2 to produce one mole of CO_2 , thus requiring 1.5 times the volume of O_2 compared to CO, leading to the conclusion that 50 liters of CO would need 75 liters of O_2 and produce 150 liters of CO_2 under the same conditions of temperature and pressure.

Long Form Response

11. **Answer:** To determine the pH of a 0.2 M NH solution, we start with the equilibrium reaction for the base ionization of ammonia: $NH + H_2O \rightleftharpoons NH_4^+ + OH^-$. The base dissociation constant, K_b , is given as 1.8×10^{-5} . Setting up the equilibrium expression, $K_b = \frac{[NH_4^+][OH^-]}{[NH]}$, and letting x represent the concentration of OH^- produced, we can express this as $K_b = \frac{(x)(x)}{(0.2 - x)}$. Assuming x is small relative to 0.2, we simplify to $K_b = \frac{x^2}{0.2}$. Solving for x gives us $x = \sqrt{K_b \times 0.2} = \sqrt{(1.8 \times 10^{-5} \times 0.2)} \approx 0.00134$ M for $[OH^-]$. The pOH is then calculated as $pOH = -\log(0.00134) \approx 2.87$, leading to a pH of $14 - pOH = 14 - 2.87 = 11.13$. Thus, the pH of the 0.2 M NH solution is approximately 11.13, confirming that the solution is basic and consistent with the expected behavior of ammonia.

Note: correct because it correctly identifies the base ionization reaction of ammonia, applies the appropriate equilibrium expression for K_b , and utilizes the assumption of negligible x to simplify the calculation for the hydroxide ion concentration, ultimately allowing for the determination of pH from the resulting concentration of hydroxide ions.

12. **Answer:** When Fehling's solution is added to diabetic urine containing sugar, a redox reaction occurs, wherein the reducing sugar, such as glucose, reduces the blue cupric ions (Cu^{2+}) present in the solution to cuprous ions (Cu). Upon heating, the cuprous ions precipitate out of solution as copper(I) oxide (CuO), which appears as a brick-red solid. This transformation is indicative of the presence of reducing sugars in the urine, serving as a qualitative test for diabetes. The underlying chemical process involves the oxidation of the sugar and the concomitant reduction of copper ions, demonstrating a classic example of redox chemistry in biological analysis.

Note: correct because it accurately describes the redox reaction between the reducing sugar in diabetic urine and the cupric ions in Fehling's solution, leading to the formation of a brick-red precipitate of copper(I) oxide, which serves as a qualitative indicator of reducing sugars and, by extension, diabetes.